

MATHEUS PAVANELI

Senior Fullstack Engineer · TypeScript · Python · Node.js / NestJS · React / Next.js · PostgreSQL · AWS · LLM Systems
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[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

English C1 — used daily in production: sprint ceremonies, code review and technical docs with international teams

SUMMARY

Fullstack engineer building production systems across EdTech, social platforms and SaaS — greenfield architecture and legacy rescue alike. I make the foundational calls: stack selection, API contracts, auth models, deployment strategy. Recent work spans a platform serving 100K+ students, a social network owned end-to-end, and a multi-provider LLM orchestration layer.

TECHNICAL SKILLS

Languages & Frameworks: TypeScript · JavaScript · Python · PHP · React · Next.js · Vue.js · Node.js · NestJS · FastAPI · Django / DRF · Fastify · Express · Laravel · TailwindCSS

AI Engineering: LLM orchestration · multi-agent routing · provider failover · response caching · model evaluation · MongoDB retrieval · LangChain · Pydantic structured output · embeddings & vector search (pgvector) · scikit-learn · NumPy

Data & Messaging: PostgreSQL · MySQL · MongoDB · DynamoDB · Redis · Firebase · Kafka · RabbitMQ · BullMQ · Celery · SQLAlchemy / Alembic · pandas

Cloud & Infrastructure: AWS (EC2, Lambda, S3, SQS, SNS, API Gateway, CloudFront, RDS, ECS, Fargate) · Azure (App Service, Functions, Service Bus, AKS) · Docker · Kubernetes · Nginx · Linux · Git · CI/CD · boto3

Architecture & APIs: Clean Architecture · DDD · Event-Driven · Microservices · CQRS · Hexagonal · Serverless · SOLID · REST · GraphQL · Jest · pytest · mypy / Ruff

PROFESSIONAL EXPERIENCE

Bernoulli Educação — EdTech · 100K+ students across Brazil

Software Engineer · Remote

Nov 2025 – Present

- Designed and delivered the company's **LLM orchestration architecture**: an orchestrator routing requests across specialized agents, **provider failover** so one vendor outage can't take the feature down, and a **MongoDB layer** for document retrieval, agent execution logs and a response cache that cuts repeat inference cost. Picked the provider by **benchmarking 5 LLMs** on cost, latency and output quality.
- Traced a core query's latency to stale planner statistics using EXPLAIN ANALYZE — the estimated vs actual row divergence exposed a sequential scan chosen over an existing index. Refreshed statistics, restored correct index usage, and tuned autovacuum thresholds so it wouldn't recur. **2s → ~150ms**, then documented the method for the team.
- Shifted releases from weekly batches to daily continuous deployment by introducing CI/CD with automated test suites — smaller changesets, cheaper rollbacks, lower risk per release.
- Authored the team's code review standard, now applied across every backend service: named the recurring anti-patterns and gave each one a concrete refactoring path.

Eicode — software consultancy · Brazilian and international clients

Senior Fullstack Engineer (promoted from mid-level) · Remote

Oct 2024 – Jun 2025

- Became sole technical owner of the company's greenfield social platform — NestJS, Next.js, PostgreSQL. **Shipped 12 feature modules** with no senior engineer above me; set the technical direction the other three engineers on the build worked to.
- Built the recommendation engine from scratch — **TF-IDF vectorization with cosine similarity** over user-generated content, served as a standalone **Python service (FastAPI, scikit-learn/NumPy)** alongside the NestJS core; **Celery workers on Redis** precompute rankings off the request path, taking feed latency **800ms → ~100ms**.
- Added semantic search with **sentence embeddings in pgvector** over **~5K indexed documents**, replacing keyword matching that missed paraphrased queries; wrote the migration and backfill tooling in **pandas**, reconciling **200K rows** against a live database with no downtime.
- Technical lead on international engagements, with **requirements, code review and stakeholder communication conducted in English**; drove Brazilian-market delivery directly alongside the Product Owner.

Jorrovi Calçados — footwear retail · internal financial and administrative systems

Junior Fullstack Engineer (promoted from trainee) · Part-time · Remote

Apr 2023 – Jun 2025

- Inherited an abandoned **financial dashboard** — revenue, cost and cash-flow reporting used from the operations floor up to the owners — with no handoff and no documentation. Reverse-engineered it, assessed the debt, and set the migration strategy alone. Promoted within 6 months.
- Stabilized it on three fronts: eliminated the crashes that made it unreliable, **traced and corrected the inconsistent figures it had been reporting**, and cut load times — restoring confidence in numbers the business was already making decisions on.
- Migrated **40+ views** to React over 15 months while the system stayed live and in daily use — incremental cutover, no feature freeze, no big-bang rewrite, no interruption to financial reporting.
- Rolled TypeScript across the full codebase after making the case to non-technical stakeholders, and automated Kubernetes deployments across dev, staging and production.

FREELANCE & CONTRACT WORK

Independent Contractor via Workana — 5+ concurrent client engagements, delivered alongside full-time roles

Senior Fullstack Engineer · Remote

Jan 2024 – Nov 2025

- Cut a frontend Docker image from **1.9 GB to ~210 MB (~89%)** by diagnosing the cause and rebuilding the pipeline with multi-stage builds, with no prior documentation to work from. Smaller artifacts dropped registry and transfer costs **over 60%**.
- Delivered Python services across **3 client platforms** — **FastAPI with Pydantic** validation and JWT auth, **~40 endpoints** total; ingestion pipelines pulling **5K records/day** from 3 sources with dedup and **idempotent Celery retries** so re-runs never duplicated writes; and a **LangChain RAG** feature over **~2K client documents** with response caching to cut cost per query.
- Built multi-provider authentication from the ground up — email, Google and Microsoft flows, 2FA, account recovery, PostgreSQL-backed hashing. **No auth-related incidents reported post-launch.**
- Shipped production Stripe billing for **3 client platforms** — subscription flows and webhooks built for idempotency, so retries never double-charged.

EDUCATION B.Eng. Software Engineering — Cruzeiro do Sul Virtual (in progress, expected 2029)